

Why Some Parameters Run and Others Cannot

Effective Drift, Standing Collapse, and Scale Transport under Admissibility Constraints

Amos Jay Maley

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Abstract

Some physical parameters vary with scale, scheme, or effective regime, while others cannot vary without changing the theory being discussed. This paper develops a structural criterion for that distinction. Building on prior fixed-domain, role-classification, and renormalization analyses of physical constants, it argues that running is not free numerical variation. Running is a licensed mobility mode: lawful scale transport preserving overlap witnesses, standing quotients, and effective envelopes. Coupling constants can run because their standing is carried by transport-defined trajectories rather than by raw numerical sameness. Effective coefficients may drift inside bounded envelopes. Scheme variables may change as presentation-dependent coordinates. Universal data may remain quotient-stable even when microscopic parameters differ. By contrast, gauge structure, object-class-defining symmetry, admissibility-enabling law form, and other anchor conditions cannot run within the same admissibility domain, because their variation changes the conditions under which objects, observables, and transport maps are defined.

The central result is a Parameter Mobility Theorem. In a scale-indexed admissibility domain, every legitimate apparent parameter change falls into one of five mobility modes: presentation change, effective drift, transport-defined running, quotient-stable invariance, or admissibility-fixed non-runability. Dimensionless status does not decide which mode obtains. Dimensionless quantities may run, remain quotient-stable, function as boundary-envelope data, encode transport, or serve as anchors depending on how standing-bearing identity is preserved. The paper does not introduce new beta functions, deny RG running, reject EFT fitting, derive numerical constants, or replace field-theoretic practice. It supplies a normal form for interpreting why some parameters run, why others only appear to change, and why still others collapse admissibility if varied inside a fixed comparison domain.

Orientation. The companion Fixity result established that standing-bearing constants are not freely variable same-domain parameters. The companion Classification result separated constant-like quantities into admissibility roles. The companion RG result treated renormalization as standing-preserving scale transport. The present paper asks the next question: given those results, which parameters are mobile, which are only presentation-dependent, and which cannot run without changing the domain?

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1 Introduction: The Run/Non-Run Problem

Physics uses the language of variation in several different ways. A gauge coupling may run with scale. A Wilson coefficient may drift under matching. A renormalization-scheme coordinate may change while observables remain fixed. A regulator may be altered without changing physical predictions. A relevant operator may control a low-energy trajectory. An irrelevant operator may disappear from the long-distance standing target. A fixed point may stop coordinate running while preserving nontrivial universal data. A gauge group, however, does not run in the same sense inside a fixed theory. Neither does the law-form by which observables, states, charges, or transport maps are defined. Those structures can be replaced by changing the theory, but that replacement is not running inside the same admissibility domain.

The difficulty is that all of these cases can be described loosely as parameter change. That loose description obscures the physical distinction between lawful mobility and domain change. The present paper develops a criterion for that distinction. It treats runnability as a role property. A parameter runs only when its change is mediated by admissible transport preserving the standing-bearing structure that makes it the same physical parameter across descriptions. When the change lacks such transport, it is not running. It is presentation change, envelope-bounded drift, quotient-stable invariance, theory change, or standing failure.

The organizing claim is:

A parameter admits genuine running only when variation is carried by admissible transport. Where variation changes merely representation, it is presentation-dependent. Where variation is valid only inside an effective regime, it is effective drift. Where variation is governed by scale transport, it is transport-defined running. Where raw differences disappear under admissible identification, the standing object is quotient-stable. Where variation alters the admissibility conditions of the domain, the parameter is non-runnable within that domain.

This claim is not a replacement for the renormalization group. It is an interpretation of what makes RG variation physically licensed. The same perspective explains why dimensionless quantities behave differently from one another. Dimensionlessness removes unit dependence, but it does not determine mobility. The fine-structure constant at a scale, a critical exponent, a mass ratio, a CKM angle, a group-theoretic factor, and a normalization ratio are all dimensionless in some sense; they do not therefore share the same admissibility role.

1.1 Why Paper 4 is needed

The previous papers in this arc established three ingredients. First, fixed-domain comparison blocks the treatment of standing-bearing constants as freely variable same-domain numbers. Second, constant-like quantities decompose into distinct roles: admissibility-fixed, quotient-dependent, effective, transport-defined, and presentation-dependent. Third, RG flow supplies the major physical

case in which scale-dependent parameter change is legitimate because standing-bearing content is transported through overlap witnesses and quotients.

Those ingredients still leave an important question open. If some quantities run and others cannot, what decides the difference? It cannot be mere numerical status, because both running couplings and fixed critical exponents are numerical. It cannot be dimensionlessness, because dimensionless quantities occupy several roles. It cannot be empirical fitting, because both envelope-bound coefficients and anchor-defining structures may be empirically constrained. The criterion must be structural: it must identify which kind of standing is preserved under variation.

The present paper therefore converts role classification into mobility classification. It asks not only what kind of parameter a quantity is, but what kinds of change are admissible for that kind of parameter.

1.2 Main contributions

The paper contributes five results.

- (C1) It defines standing-bearing parameter identity for scale-indexed admissibility domains.
- (C2) It proves a Parameter Mobility Theorem separating five mobility modes: presentation change, effective drift, transport-defined running, quotient-stable invariance, and admissibility-fixed non-runnability.
- (C3) It explains why dimensionless status does not determine mobility.
- (C4) It distinguishes coupling running from anchor variation by contrasting gauge couplings with gauge structure.
- (C5) It supplies operational criteria for effective drift, fixed points, universality, operator relevance, threshold matching, no-default mobility, and null endpoint exclusion.

1.3 Non-claims

The analysis does not claim:

- (N1) new beta functions or a new RG formalism;
- (N2) numerical derivation of physical constants;
- (N3) denial that couplings run;
- (N4) denial of scheme dependence;
- (N5) denial of regulator or cutoff methods;
- (N6) rejection of Wilsonian or perturbative EFT;
- (N7) proof of a UV completion;
- (N8) that all EFT coefficients are internally derivable;
- (N9) that BSM models are inadmissible;
- (N10) that parameter scans lack heuristic or phenomenological value.

The claim is narrower: mobility is admissible only when the relevant identity conditions are preserved. The paper classifies those conditions.

2 What This Paper Adds to Standard Running Language

Standard RG and EFT practice already contain the formal machinery used in this paper: beta functions, anomalous dimensions, matching equations, relevant and irrelevant operators, threshold corrections, and fixed-point analysis. The present analysis does not replace that machinery. Its contribution is to classify the standing role of the quantities that appear in it. It asks a different question from the one answered by a beta-function calculation. A beta function tells us how a coordinate changes with scale. The mobility analysis asks what must be preserved for that change to count as running of the same physical parameter rather than presentation change, envelope drift, quotient identification, or domain replacement.

The distinction is useful because the same notation often hides different mobility roles. A parameter symbol may denote a scheme coordinate in one calculation, an empirically fitted EFT coefficient in another, a boundary input in a third, and a quotient-stable invariant in a fourth. Without role discipline, one may infer too much from the fact that all of these objects can be written as numbers. The mobility theorem supplies a normal form: before asking whether a parameter runs, one first asks what carries its identity across comparison.

The paper also clarifies what is not at issue. No claim is made that established RG practice lacks rigor. No claim is made that conventional terminology should be abandoned. The claim is that physical interpretation of parameter mobility requires an additional layer of role classification. This layer becomes important whenever running couplings are used in arguments about constant variation, fine-tuning, UV completion, effective theory validity, universality, or model comparison.

Three questions are kept separate throughout:

- (Q1) **How does the value change?** This is the formal RG, matching, or redescription question.
 - (Q2) **What preserves physical identity?** This is the witness, quotient, and transport question.
 - (Q3) **What permits the comparison?** This is the admissibility-domain and envelope question.
- A quantity genuinely runs only when all three questions receive compatible answers.

3 Fixed Domains, Scale Transport, and Parameter Identity

3.1 Fixed admissibility domains

A comparison of parameters requires a domain in which physical identity is stable. The relevant object is not a bare equation containing symbols, but a structured physical setting in which observables, transformations, witnesses, and reference relations are fixed enough for comparison.

Definition 3.1 (Fixed admissibility domain). *A fixed admissibility domain is a tuple*

$$D = (\mathcal{T}, \mathcal{A}, \mathcal{W}, \sim, \mathcal{E})$$

where $\mathcal{T}$ is the theoretical presentation, $\mathcal{A}$ is the admissibility envelope, $\mathcal{W}$ is the witness ledger of admissible observables and empirical reconstruction relations, $\sim$ is the standing quotient identifying descriptions with the same physical content, and $\mathcal{E}$ is the operative regime or envelope within which the description is licensed. A same-domain comparison preserves these components up to admissible redescription.

This definition does not require all microscopic information to be retained. It requires that the comparison target be explicit. A low-energy EFT may discard high-energy details while retaining a valid domain. A universality class may identify many microscopic systems under a standing quotient. The point is not maximal information retention. The point is controlled identity.

3.2 Scale-indexed domains

RG analysis introduces scale-indexed domains. At each scale, the description may carry different coefficients, fields, regulators, or effective variables. The same physical content is not preserved by numerical equality of all entries. It is preserved by lawful scale transport.

Definition 3.2 (Scale-indexed admissibility domain). *A scale-indexed admissibility domain is a family*

$$\mathcal{D}_\mu = (\mathcal{T}_\mu, \mathcal{A}_\mu, \mathcal{W}_\mu, \sim_\mu, \mathcal{E}_\mu, \mathcal{R}_{\mu \rightarrow \nu})$$

where $\mathcal{R}_{\mu \rightarrow \nu}$ is a scale-transport map from the description at scale μ to the description at scale ν . The map is admissible when it preserves overlap witness structure, factors through the standing quotients, and respects the relevant envelopes or matching data.

Definition 3.3 (Overlap witness structure). *Two scale-indexed descriptions have overlap witness structure when their admissible observables agree on the common reconstruction domain after any required coarse-graining, rescaling, field redefinition, or matching operation. Overlap witnesses are not all observables of either description; they are the observables that both descriptions can lawfully compare.*

Overlap witnesses make scale comparison physical rather than merely formal. Without them, a changed parameter may be a different coordinate, a different theory, or an unlicensed extrapolation. With them, RG flow can carry identity through changing descriptions.

3.3 Standing-bearing parameter identity

Definition 3.4 (Standing-bearing parameter identity). *A parameter-like quantity p has standing-bearing identity in a scale-indexed domain when its use participates in stable empirical reconstruction, lawful transport, quotient identity, admissibility-envelope continuity, or witness-preserving comparison. A symbol or numerical value lacking these relations is not yet a physical parameter identity.*

The relevant question is therefore not simply whether a value changes. The relevant question is what kind of identity survives the change. A raw symbol can change with no physical difference. A coordinate can run while the transport class remains fixed. A quotient object can remain invariant while microscopic presentations differ. An anchor can be changed only by changing the domain.

Definition 3.5 (Parameter running). *A parameter $p(\mu)$ genuinely runs from μ to ν only if there exists an admissible transport map $\mathcal{R}_{\mu \rightarrow \nu}$ such that the change in p is part of a witness-preserving, quotient-respecting, envelope-compatible scale continuation. Raw numerical change without this transport is not running in the physical sense used here.*

This definition is compatible with ordinary RG equations. For a coupling,

$$\mu \frac{dg}{d\mu} = \beta(g),$$

the differential equation gives a transport rule for a scale-indexed coordinate. Physical standing enters when the flow is tied to overlap observables, matching conditions, and quotient-preserving comparison.

4 Five Modes of Parameter Mobility

The mobility of a parameter is determined by what kind of standing is preserved under its apparent variation. Five primitive modes exhaust the available admissible cases.

4.1 Mobility modes

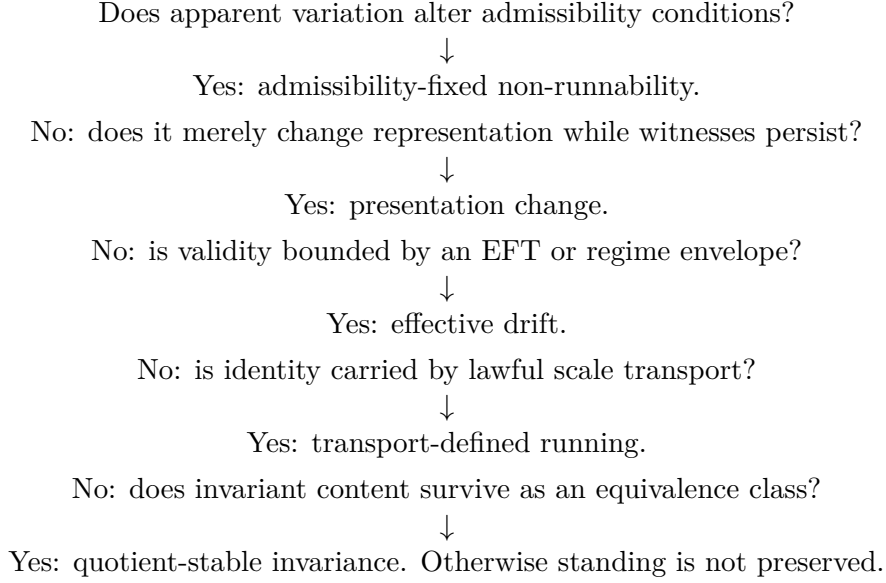
Table 1: Five mobility modes for parameter-like quantities.

Mode	What changes	What preserves standing	Typical examples
Presentation change	Units, coordinates, scheme labels, field normalizations, regulator representation	Witness structure and quotient identity are unchanged	Unit conventions, renormalization coordinates, field redefinitions
Effective drift	Coefficients or fitted parameters inside a bounded envelope	Envelope validity, matching rules, and overlap witnesses	Wilson coefficients, low-energy constants, phenomenological EFT parameters
Transport-defined running	Scale-indexed coordinate changes by a lawful flow	RG trajectory, beta-function transport, overlap observables	Gauge couplings, running masses, anomalous dimensions in context
Quotient-stable invariance	Microscopic parameters differ or flow away, but quotient content remains	Universality class, fixed-point data, critical exponents, invariant combinations	Critical exponents, universal amplitude ratios, transport-stable fixed-point data
Admissibility-fixed non-runability	Attempted change alters domain eligibility	Object-class, law-form, admissible observables, charge algebra, transport maps	Gauge group within fixed theory, representation class, causal/signature anchor, law-form constraints

The modes should not be confused with degrees of reality. An effective parameter can be physically real inside its envelope. A presentation-dependent variable can be indispensable for calculation. A quotient-stable invariant may be less microscopic but more robust for the target question. An anchor may not be numerical, yet it can determine whether numerical parameters are meaningful.

4.2 Decision flow

A compact decision flow is useful:



This flow is not a calculation of beta functions. It is a classification of what a beta function, coefficient, invariant, or anchor is doing in physical comparison.

4.3 Presentation change

Lemma 4.1 (Presentation Change Lemma). *If a parameter-like value changes under an admissible redescription while all overlap witnesses and standing quotients are preserved, then the change is presentation-dependent rather than physical running.*

Proof. By hypothesis the witness ledger and standing quotient are unchanged. The value change therefore does not alter empirical reconstruction or physical identity in the comparison domain. Its mobility lies in representation, not in standing-bearing content. $\square$

Examples include units, coordinate charts, field normalizations, and scheme labels when those choices leave physical comparisons invariant. Presentation dependence is not dismissal. It identifies a quantity's role: it organizes representation rather than independently carrying the physical identity target.

4.4 Effective drift

Lemma 4.2 (Effective Drift Lemma). *If a parameter-like value changes inside a bounded regime and its standing is licensed only by that regime's envelope, matching rules, and witness ledger, then the change is effective drift.*

Proof. The parameter has physical use only relative to $\mathcal{E}$. Variation outside $\mathcal{E}$ is not automatically a continuation of the same quantity. Within $\mathcal{E}$, drift is admissible when witnesses and matching relations preserve empirical reconstruction. Thus the mobility is real but envelope-bound. $\square$

Effective drift is common in EFT. A Wilson coefficient may change with scale or matching convention while remaining meaningful within the effective domain. This is not inadequacy. Envelope-bounded validity specifies the conditions under which the coefficient preserves standing.

4.5 Transport-defined running

Lemma 4.3 (Transport-Defined Running Lemma). *A parameter genuinely runs when its value change is generated by an admissible transport map $\mathcal{R}_{\mu \rightarrow \nu}$ that preserves overlap witness structure and factors through standing quotients.*

Proof. Running is not defined by numerical difference alone. It requires a transport law identifying how the parameter at one scale reconstructs the parameter at another. If the transport law preserves overlap witnesses and quotient identity, the parameter's standing is carried by the flow. The change is therefore genuine running rather than arbitrary reassignment. $\square$

Gauge couplings provide the paradigm. Their raw values are scale- and scheme-indexed, but the RG trajectory and its witness relations carry physical identity.

4.6 Quotient-stable invariance

Lemma 4.4 (Quotient-Stable Invariance Lemma). *If microscopic or coordinate-level differences fail to survive admissible quotienting while invariant physical content remains preserved, the standing object is quotient-stable rather than running as a raw value.*

Proof. The differences are present before quotienting, but the physical comparison target is the equivalence class selected by admissible witnesses and transport. Since the quotient object, not the raw microscopic data, is the standing target, its persistence is quotient-stable invariance. $\square$

Universality classes illustrate this mode. Many microscopic Hamiltonians or Lagrangians may flow toward the same long-distance standing target. The individual microscopic parameters have not become numerically identical. Their differences have ceased to be part of the quotient target relevant to the universality question.

4.7 Admissibility-fixed non-runability

Lemma 4.5 (Anchor Non-Runability Lemma). *If varying a parameter-like structure changes the admissibility conditions under which objects, observables, quotient identity, or transport maps are defined, then the structure is non-runable within that domain.*

Proof. Running presupposes a comparison domain in which the parameter can be identified across scales. If the attempted variation changes the domain's admissibility conditions, then the identity relation required for same-domain running is lost. The operation may define a different theory or envelope, but it is not running within the original domain. $\square$

This is the central distinction between gauge couplings and gauge structure. Couplings may be coordinates on a transport trajectory. Gauge structure fixes the charge algebra, representation content, and admissible field relations of the comparison domain. Changing it is not a running of the same parameter; it is a change of domain.

5 The Parameter Mobility Theorem

Definition 5.1 (Mobility profile). *The mobility profile of a parameter-like quantity p in a scale-indexed admissibility domain is*

$$\text{Prof}(p) \in \{P, E, T, Q, A\},$$

where P denotes presentation change, E effective drift, T transport-defined running, Q quotient-stable invariance, and A admissibility-fixed non-runnability. Composite quantities may have composite profiles, but each primitive admissibility-bearing contribution belongs to one of these loci.

Theorem 5.2 (Parameter Mobility Theorem). *Let p be a parameter-like quantity in a scale-indexed admissibility domain D_μ . Any admissible apparent change of p falls into exactly one primitive mobility mode: presentation change, effective drift, transport-defined running, quotient-stable invariance, or admissibility-fixed non-runnability. Genuine running occurs only in the transport-defined case, possibly combined with effective envelope dependence. If apparent variation preserves only presentation, it is not running; if it preserves quotient identity while raw differences disappear, the standing object is quotient-stable; if it changes admissibility conditions, it is not same-domain running but domain change or standing failure.*

Proof. Consider an apparent change of p . If the change affects only coordinates, units, field normalization, scheme labels, or other representational choices while overlap witnesses and standing quotients are preserved, the Presentation Change Lemma applies. If the change is licensed only relative to a bounded envelope, matching rules, and witness ledger, the Effective Drift Lemma applies. If the change is mediated by an RG or other lawful transport map that preserves overlap witnesses and factors through the standing quotient, the Transport-Defined Running Lemma applies. If raw or microscopic differences are eliminated by admissible quotienting while invariant content remains, the Quotient-Stable Invariance Lemma applies. If attempted change alters the admissibility conditions that define objects, observables, quotient identity, or transport maps, the Anchor Non-Runnability Lemma applies.

These alternatives exhaust the available loci of standing-bearing comparison. Apparent variation must either preserve presentation only, remain envelope-bound, be transported lawfully, survive as quotient identity, or alter the domain conditions. A proposed sixth primitive mode would need to change none of presentation, envelope, transport, quotient, or admissibility while still changing the physical parameter. Such a change has no remaining standing-bearing locus in the comparison domain. It is therefore either a composite of the five modes, an unspecified selector, or a failure of parameter identity. $\square$

Corollary 5.3 (Running is not free variation). *A parameter can run only when its mobility is transport-defined. Numerical variability, dimensionlessness, empirical fitting, or formal appearance in an equation is insufficient for runnability.*

Corollary 5.4 (Non-runnability is not metaphysical rigidity). *If a structure cannot run within a fixed admissibility domain, it may still differ across theories, models, UV completions, or envelopes. The claim is same-domain non-runnability, not global metaphysical immutability.*

This theorem sharpens the comparison between couplings and anchors. The fact that a coupling runs does not imply that every parameter-like structure is mobile. The fact that an anchor cannot run does not imply that all numerical constants are fixed in raw value. Mobility is role-dependent.

5.1 Composite mobility profiles

Many quantities encountered in physics have composite profiles. A running mass may be an effective coefficient in one envelope, a transport-defined coordinate under RG, and a boundary-fitted input relative to a lower-energy measurement. A Yukawa parameter may run under RG while its low-energy numerical value remains an empirical input. A Wilson coefficient may be both an effective drift quantity and a matching residue. The mobility theorem is not invalidated by such composites; it requires that each primitive contribution be separated before explanatory claims are made.

Proposition 5.5 (Composite Profile Decomposition). *If a parameter-like symbol carries several admissibility roles, its apparent mobility decomposes into primitive presentation, effective, transport, quotient, or anchor components. No composite profile licenses a sixth primitive mobility mode.*

Proof. A composite symbol can participate in several relations at once. For each relation, the preserved structure is either representational, envelope-bound, transport-mediated, quotient-stable, or domain-fixing. These are the primitive standing loci exhausted in Theorem 5.2. The composite has several roles, but the roles are drawn from the same five-mode set. $\square$

This decomposition is particularly important for Standard-Model parameters. Gauge couplings run. Yukawa matrices run. Mixing parameters are reconstructed from basis relations and may inherit scale dependence through the running of the underlying matrices. Mass ratios can be quotient-dependent while also envelope-sensitive. None of these facts make all entries members of one homogeneous mobility class.

6 Why Dimensionless Structure Behaves Differently

Dimensionless quantities are often treated as the cleanest cases of physical constants because unit arbitrariness has been removed. This is correct as far as it goes, but incomplete. Dimensionlessness removes one source of presentation dependence; it does not determine admissibility role or mobility profile.

Proposition 6.1 (Dimensionlessness Non-Determination Proposition). *Dimensionless status alone determines neither fixity nor mobility. A dimensionless quantity may be presentation-dependent, effective, transport-defined, quotient-stable, or admissibility-fixed depending on how its standing-bearing identity is preserved.*

Proof. A dimensionless quantity can be a pure normalization ratio, a scale-indexed effective coupling, a critical exponent, a mass ratio, a mixing angle, a CP phase, or a group-theoretic invariant. These cases preserve identity through different structures: presentation convention, envelope validity, transport, quotient persistence, or domain anchoring. Since the same dimensionless status is compatible with all five mobility modes, dimensionlessness does not determine the mode. $\square$

Table 2: Dimensionless quantities and mobility roles.

Quantity type	Mobility profile	Rationale
Fine-structure constant $\alpha(\mu)$	Effective / transport-defined	Its numerical value is scale-indexed and scheme-contextual; standing is carried by transport and observables.
Critical exponent	Quotient-stable	It is universal data preserved across microscopic differences within a universality class.
Mass ratio	Quotient-dependent / envelope-bound	Unit independence is real, but physical role depends on the mass-generating domain and comparison envelope.

Quantity type	Mobility profile	Rationale
CKM or PMNS angle	Transport/reconstruction-defined	It encodes reconstruction between admissible bases, not an independent raw scalar.
CP phase residue	Transport-defined tensor content	Its standing appears through observable interference or asymmetry under lawful reconstruction.
Group-theoretic factor	Admissibility-fixed or quotient-stable	Its role depends on whether it fixes representation structure or survives quotient classification.
Normalization ratio	Presentation-dependent	It may be dimensionless while changing only conventions.

The proposition explains why dimensionless constants are simultaneously important and heterogeneous. Unit discipline makes them better candidates for invariant physical content, but it does not settle whether they run, remain fixed, serve as boundary data, or encode transport. Their mobility must be classified by standing preservation.

6.1 Dimensionless running and dimensionless stability

A dimensionless coupling can run because scale is supplied externally by the RG comparison. Its dimensionlessness does not remove scale dependence. Conversely, a critical exponent can be dimensionless and quotient-stable because it is extracted from a fixed point or universality class. A CKM angle is dimensionless but reconstruction-defined: its physical meaning depends on the relation between weak and mass bases. A group factor can be dimensionless but anchor-like when it fixes the representation algebra of the domain.

The physically meaningful distinction is not dimensionless versus dimensionful. It is how the quantity carries standing.

7 Why Coupling Constants Can Run

Gauge couplings are the central case. Their scale dependence is not a failure of constant identity. It is the paradigm of transport-defined mobility.

7.1 Raw coupling labels and transport classes

A coupling written as g is incomplete without scale, scheme, normalization, and witness context. Under RG flow,

$$\mu \frac{dg}{d\mu} = \beta(g),$$

the value $g(\mu)$ changes. The standing-bearing object is not raw numerical sameness across μ . It is the transport relation by which scale-indexed descriptions reconstruct overlap observables.

Proposition 7.1 (Coupling Running Proposition). *A coupling constant can run when its identity is carried by a scale-transport class rather than by raw numerical equality. The physical content lies*

in the admissible relation among $g(\mu)$, the beta function, scheme discipline, matching data, and overlap witnesses.

Proof. The coupling value at a single scale is not a complete invariant. The RG equation supplies a lawful relation between values at different scales. If the relation preserves overlap observables and quotient identity, the coupling's standing is transported through the trajectory. Thus the coupling can run without becoming an arbitrary free scalar. $\square$

What changes	What survives
Raw coordinate value $g(\mu)$	RG transport class and observable reconstruction
Scheme representation	Scheme-independent overlap observables and universal data where applicable
Effective coefficients	Envelope-licensed predictions and matching relations
Field normalization	Physical amplitudes, cross sections, or scaling quantities after admissible reconstruction
Regulator details	Renormalized witness structure when the regulator is not promoted beyond its role

7.2 Dimensional transmutation

Dimensional transmutation is an instructive mobility conversion. A dimensionless running coupling can be traded for a scale such as Λ_{QCD} under appropriate RG conditions. This is not derivation from nothing. It is a change in role profile: a scale-indexed coupling coordinate is converted into a transmutation scale that functions as an effective anchor for a low-energy envelope.

Proposition 7.2 (Transmutation Role Conversion). *When dimensional transmutation replaces a running coupling coordinate by a scale parameter, the mobility profile changes from transport-defined coordinate to effective scale anchor. The resulting scale has standing only through the RG trajectory, boundary input, and witness structure that define it.*

The proposition avoids two errors. It does not treat Λ_{QCD} as an arbitrary raw constant. It also does not claim that its numerical value is internally forced without boundary data. Its standing is inherited through transport and empirical fixation.

8 Why Some Constants Cannot Run

The fact that couplings can run does not generalize to all parameter-like structures. Some structures make running possible by fixing the domain in which running is defined. These cannot run within that same domain.

8.1 Anchors and domain change

An anchor is a structure that fixes the admissibility conditions of a domain: which objects exist in the theory, which transformations are lawful, which observables are admissible, and which witness comparisons are meaningful. Gauge group, representation class, causal/signature structure, and law-form constraints can function this way.

Proposition 8.1 (Gauge-Structure Non-Running Proposition). *Within a fixed gauge theory, the gauge group and representation-defining structure do not run in the same sense as gauge couplings. Changing them changes the charge algebra, field content, admissible observables, and transport maps of the domain.*

Proof. A running coupling presupposes a fixed comparison framework in which the coupling can be identified at different scales. The gauge structure supplies part of that framework: it determines representations, charges, covariant derivatives, allowed interactions, and the form of the beta-function analysis. If that structure is varied, the identity conditions for the coupling and witnesses change. The operation may define a new theory or phase description, but it is not same-domain running of the gauge structure. $\square$

This does not deny model building in which gauge symmetry differs at high energy or is broken at low energy. Symmetry breaking, unification, and BSM structure may be physically meaningful. The point is that such changes require matching, embedding, phase structure, or domain transition. They are not raw running of the same anchor inside an unchanged domain.

8.2 Object-class-defining symmetries

The same logic applies to object-class-defining symmetries. A parameter that fixes what counts as a state, field, charge, observable, or allowed transformation cannot be varied without changing the identity relation used for comparison. It may be replaced by a larger structure, broken by a vacuum choice, or realized differently in another phase. But its same-domain runnability is not analogous to a coupling's beta-function evolution.

Corollary 8.2 (Anchor variation is domain transition). *Variation of an admissibility anchor is a domain transition unless a higher-level admissible transport map embeds both anchor states in a common comparison domain.*

This corollary is important for grand unification or symmetry-breaking scenarios. A larger theory may contain multiple effective domains with different apparent anchors. In that case, the mobility belongs to the embedding and matching structure, not to unconstrained running of the lower-domain anchor.

8.3 Compatibility with symmetry breaking and unification

The non-runability of anchors within a fixed domain is compatible with symmetry breaking, dual descriptions, emergent gauge structure, and high-energy unification. Those cases are not raw same-domain anchor running. They require an enlarged comparison setting in which the lower-domain structures are related by embedding, phase transition, matching, or duality. Once that enlarged setting is supplied, the mobility claim shifts to the transport or boundary structure connecting the domains.

For example, a low-energy gauge group may arise after spontaneous symmetry breaking from a larger high-energy group. The lower-energy group did not run as an isolated anchor. It was selected through a vacuum structure and inherited standing through the embedding and matching conditions of the larger theory. Similarly, an effective gauge structure may emerge in a condensed-matter system. Its standing is then envelope-bound and witness-dependent; it is not a freely mobile anchor in the original microscopic domain.

This distinction protects both sides of the argument. It prevents anchor non-runability from becoming a denial of actual physical transitions, and it prevents physical transitions from being misclassified as unconstrained parameter mobility.

9 Effective Drift versus True Running

Effective drift and transport-defined running are closely related but not identical. The difference matters for EFT coefficients, fitted parameters, and phenomenological scans.

9.1 Effective drift

An effective coefficient has standing relative to an envelope. It may change with scale, matching convention, truncation order, or data fitting procedure. This mobility is not automatically a beta-function running of a primitive constant. It is envelope-bound.

Proposition 9.1 (Envelope Mobility Criterion). *An effective parameter admits drift only relative to an operative envelope $\mathcal{E}$, a witness ledger $\mathcal{W}$, and matching or fitting rules that preserve empirical reconstruction. Promotion of that drift outside $\mathcal{E}$ requires additional admissible transport.*

Proof. The parameter’s use is defined by the effective description. Its coefficients, truncation, and observables are not licensed outside the relevant domain without matching or completion. Thus its mobility is real within the envelope but not primitive beyond it. $\square$

9.2 True transport-defined running

True running requires a lawful scale-transport equation or map. It is not merely that fitted values change when a model is adjusted. A parameter runs when there is a rule that reconstructs the parameter across scales while preserving overlap witnesses.

The distinction can be summarized as follows:

Feature	Effective drift	Transport-defined running
License	EFT envelope, matching, fitting domain	RG map or scale-transport equation
Identity carrier	Envelope-relative predictive role	Trajectory preserving overlap witnesses
Failure mode	Scope overextension	Loss of transport or quotient preservation
Typical quantity	Wilson coefficient in a truncated EFT	Gauge coupling under beta-function evolution

The two modes may combine. A running coupling inside an EFT is both effective and transport-defined. The mobility theorem allows composite profiles, but it insists that each contribution be classified.

10 Fixed Points and Universality

10.1 Fixed points: when running stops without arbitrariness

A fixed point satisfies

$$\beta(g_*) = 0.$$

As a coordinate statement, this says that a chosen coupling coordinate does not change under the flow. Structurally, the more important claim is that the point can define a transport-stable quotient locus.

Definition 10.1 (Transport-stable quotient locus). *A point, surface, or set in coupling space is a transport-stable quotient locus when RG evolution preserves the standing quotient and admits invariant overlap observables at that locus.*

Proposition 10.2 (Fixed-Point Mobility Proposition). *A fixed point is physically standing-bearing when it functions as a transport-stable quotient locus. Its role is not arbitrary endpoint selection but scale-stable preservation of quotient identity.*

Gaussian fixed points, interacting fixed points, IR attractors, and UV candidates differ in dynamical and physical detail. The mobility classification is neutral among them. It says only that fixed-point standing depends on preserved quotient structure, not on the formal vanishing of a beta function alone.

10.2 Universality: when different parameters become the same standing object

Universality occurs when distinct microscopic presentations flow into the same long-distance standing target. It is not that microscopic parameters literally become equal. Rather, their differences fail to survive the quotient relevant to the long-distance witness structure.

Theorem 10.3 (Universality Mobility Theorem). *Microscopic parameter differences become quotient-equivalent under RG when admissible coarse-graining maps preserve the same overlap witness structure and factor through the same standing quotient. In that case, the universal data are quotient-stable, while the microscopic differences are not part of the mobility target.*

Proof. Let two microscopic descriptions be connected to long-distance descriptions by admissible coarse-graining. If the resulting long-distance descriptions agree on overlap witnesses and identify under the same quotient, then the microscopic differences do not alter the standing object relevant to the universality class. The universal quantities persist as quotient-stable invariants. The microscopic differences may remain real in a different envelope, but they are not standing-bearing for this quotient target. $\square$

This theorem clarifies how some parameters can “run out” of physical relevance without becoming nonexistent. Irrelevant microscopic details are not denied. They are outside the quotient target of the universality problem.

11 Operators as Mobility Diagnostics

Operator relevance provides a physics-facing diagnostic for mobility.

Table 3: Operator relevance as a mobility diagnostic.

Operator type	Mobility role	Admissibility interpretation
Relevant	Effective trajectory control / boundary input	Alters the long-distance envelope and may require empirical or boundary fixation.
Irrelevant	Quotient-decaying contribution	Loses standing-bearing influence for the low-energy quotient, though it may matter in another envelope.

Operator type	Mobility role	Admissibility interpretation
Marginal	Requires further transport analysis	May remain fixed, run logarithmically, become marginally relevant, or become marginally irrelevant depending on beta structure.
Dangerously irrelevant	Not presentation-dependent if witnesses are affected	Cannot be discarded when it affects overlap observables, phases, or reconstruction.

Proposition 11.1 (Operator Mobility Diagnostic). *The RG relevance of an operator is a diagnostic of mobility role: it indicates whether the associated parameter controls envelope trajectory, decays under quotienting, requires transport analysis, or remains witness-bearing despite naive irrelevance.*

This proposition does not redefine operator relevance. It interprets its admissibility role. Relevance is not merely a power-counting label; in physical use it indicates how a coefficient participates in standing preservation across scale.

12 Threshold Matching and Boundary Trace

Running across thresholds is not automatic. When a heavy field is integrated out or an EFT changes regime, the low-energy description inherits standing only through matching data.

Definition 12.1 (Boundary trace). *A boundary trace is the overlap information at a regime transition that permits the downstream effective description to reconstruct the upstream witness content relevant to its envelope.*

Proposition 12.2 (Threshold Mobility Criterion). *Cross-threshold parameter mobility is admissible only when matching supplies a boundary trace preserving overlap witness structure between the pre-threshold and post-threshold descriptions.*

Proof. A threshold changes the degrees of freedom or effective variables available in the description. Without matching, a downstream coefficient may be numerically assigned but lacks a lawful relation to the upstream witness ledger. Matching supplies the boundary trace by which overlap observables are reconstructed. Therefore cross-threshold mobility requires boundary-trace fixation. $\square$

Consider a heavy particle integrated out below mass scale M . Above M , the theory includes the heavy field explicitly. Below M , the heavy field is absent and its effects appear in Wilson coefficients. The coefficients are not arbitrary replacements for the original dynamics. They inherit standing through matching conditions that preserve overlap observables at the boundary. Their subsequent drift is effective and transport-defined inside the low-energy envelope.

This section connects the mobility theorem to the forced-continuation material without overextending it. The point is not that thresholds force existence. It is that continuity across a threshold requires non-vacuous boundary data.

13 No Default Parameter Mobility

Before admissibility structure is fixed, there is no canonical answer to whether a parameter may run. Runnability is not a default property of a symbol. It depends on transport, witness preservation, quotient identity, envelope role, and anchor status.

Proposition 13.1 (No Default Mobility Proposition). *No pre-admissible selector canonically assigns a parameter to be running, fixed, effective, quotient-stable, or presentation-dependent before the comparison domain and standing conditions are specified.*

This proposition is deliberately narrow. It does not prohibit exploratory scans, phenomenological parameter studies, or model construction. It blocks only the promotion of an unlicensed mobility assignment into a canonical physical explanation. There is no default scale selector, endpoint selector, regulator ontology, fixed-point privilege, parameter-space measure, or mobility profile without admitted structure.

13.1 Null endpoints and collapse cases

A proposed running trajectory may fail by terminating in a null regime: no witness preservation, no quotient identity, no envelope, no transport map, no boundary trace, or only presentation without standing-bearing content.

Corollary 13.2 (Null Endpoint Exclusion). *A parameter cannot be said to run into a physically admissible endpoint unless standing-bearing structure survives the endpoint through witnesses, quotient identity, transport, or boundary trace.*

This corollary does not deny formal endpoints in equations. It says that a formal endpoint is not automatically a physical completion. The endpoint must carry enough structure to function as an admissible object of comparison.

14 Worked Examples

14.1 Gauge coupling

For a gauge coupling,

$$\mu \frac{dg}{d\mu} = \beta(g),$$

the mobility profile is transport-defined and often effective. The coordinate $g(\mu)$ changes with scale. The beta function supplies the transport rule. The scheme supplies a presentation. Thresholds supply matching data. Observables supply witnesses. Fixed points, if present, supply quotient-stable loci. Dimensional transmutation may convert the coupling trajectory into a scale anchor such as Λ_{QCD} .

Object	Mobility role	Reason
$g(\mu)$	Transport-defined coordinate	Its identity is scale-indexed and carried by RG flow.
Beta function	Transport rule	It specifies admissible scale continuation.
Scheme choice	Presentation-dependent	It changes coordinates when overlap observables are preserved.
Wilsonian cutoff	Envelope marker or regulator role	Its physical status depends on whether it marks validity or is merely calculational.
Fixed point g_*	Quotient-stable locus	It preserves standing quotient under RG evolution when witnesses persist.
Λ_{QCD}	Transmutation scale / effective anchor	It inherits standing through transport and boundary fixation.
Cross-section overlap data	Witness structure	It supplies empirical reconstruction across scale.

The conclusion is precise: gauge couplings run because their identity is transport-defined. They are not freely reassigned at each scale.

14.2 Gauge group

A gauge group G inside a fixed theory occupies a different role. It determines representation content, charge structure, covariant derivative form, allowed interactions, anomaly constraints, and the observable algebra. Changing G changes the conditions under which couplings, fields, and witnesses are defined.

Thus G is not a running parameter inside the fixed domain. A high-energy embedding, symmetry-breaking pattern, or phase transition may relate different effective structures, but that relation requires a higher-level transport or matching framework. It is not same-domain running of G .

14.3 Mass ratios, Yukawas, and mixing parameters

Mass ratios, Yukawa entries, CKM parameters, PMNS parameters, and CP phases occupy mixed profiles.

Table 4: Mobility profiles for mass and mixing quantities.

Quantity	Mobility profile	Interpretation
Mass ratio	Quotient-dependent / envelope-bound	Unit-independent but tied to mass-generation domain and comparison envelope.
Yukawa eigenvalue	Boundary-envelope input / possibly running coordinate	May run under RG, but its measured low-energy value is not thereby internally derived.
CKM angle	Reconstruction-transport parameter	Encodes relation between weak and mass bases.
PMNS angle	Reconstruction-transport parameter	Encodes oscillation reconstruction between flavor and mass descriptions.
CP phase	Transport-defined tensor residue	Has standing through observable interference or asymmetry.

The main lesson is that not every change in a numerical entry is running. Some changes are fitting, some are boundary input, some are basis reconstruction, some are effective drift, and some are domain change.

15 Error Taxonomy

The mobility theorem identifies common errors in discussions of constants and RG.

- (E1) **Scheme drift mistaken for physical running:** a coordinate changes while witnesses are preserved.
- (E2) **Effective drift promoted to unrestricted mobility:** an envelope-bound coefficient is extrapolated beyond its admissible domain.
- (E3) **Anchor change mistaken for parameter running:** a gauge group, law-form, or object-class condition is varied as if it were a coupling.
- (E4) **Dimensionlessness mistaken for primitive runnability:** unit independence is treated as mobility license.
- (E5) **Fixed points treated as arbitrary endpoints:** a beta-function zero is promoted without witness or quotient analysis.
- (E6) **Universality mistaken for numerical coincidence:** quotient persistence is reduced to accidental convergence.
- (E7) **Threshold crossing treated as automatic continuation:** matching and boundary trace are omitted.
- (E8) **Null endpoints treated as physical completions:** formal termination is accepted without standing-bearing structure.

(E9) **All numerical symbols placed in one parameter space:** mobility roles are collapsed into homogeneous variation.

These errors are not merely semantic. They change what counts as explanation. A parameter scan, for example, may be useful for exploration, but it is not a canonical account of runnability unless it specifies transport, witnesses, quotient structure, envelope, and anchor conditions.

16 Objection Matrix

Objection 1: Couplings run, so constants are not fixed.

Couplings run because their identity is transport-defined. This supports the mobility theorem rather than refuting fixed-domain discipline. Running is lawful scale transport, not free same-domain reassignment.

Objection 2: Dimensionless constants can vary.

Some dimensionless quantities can vary, some remain quotient-stable, some are presentation-dependent, and some function as anchors. Dimensionlessness removes unit dependence but does not assign a mobility profile.

Objection 3: Gauge groups can differ in BSM theories.

Yes. The claim is not that gauge structure cannot differ across theories. It is that changing gauge structure changes the admissibility domain unless a higher-level embedding or matching structure supplies lawful transport.

Objection 4: EFT parameters are fitted, so they are free.

Fitting is not unrestricted mobility. EFT coefficients have standing through an envelope, a witness ledger, and matching or calibration rules. Their values may be empirical inputs without being freely mobile primitives.

Objection 5: Fixed points are just beta-function zeros.

A beta-function zero is the formal condition. The physical standing of a fixed point depends on whether it preserves quotient identity and overlap witnesses under scale transport.

Objection 6: Universality is already standard.

The paper does not replace standard universality theory. It classifies the admissibility role of universality: universal data are quotient-stable standing structures under coarse-graining.

Objection 7: This adds no new dynamics.

Correct. The contribution is not dynamical novelty. It is a structural criterion for runnability and non-runability, clarifying when parameter variation is physically licensed.

Objection 8: This is semantic.

The classification affects what counts as a same-domain parameter, a running quantity, an effective coefficient, a quotient invariant, or a domain change. These distinctions constrain explanation and model comparison.

Objection 9: Parameter scans are useful.

Yes. The paper does not reject scans. It denies only that a scan by itself supplies canonical mobility, measure, or explanatory authority without admissibility structure.

17 Conclusion

Some parameters run because their identity is transport-defined. Some drift because their standing is effective and envelope-bound. Some change only as presentation. Some persist as quotient-stable structures even when microscopic details differ. Some cannot run inside a fixed domain because their variation changes the conditions under which observables, objects, and transport maps are defined.

The distinction is not fixed versus variable numbers. It is the distinction among admissible transport, effective drift, quotient persistence, presentation change, and domain change. That distinction explains why coupling constants can run while gauge structure cannot run in the same sense; why dimensionless status is insufficient to determine mobility; why EFT coefficients are real but envelope-bound; why fixed points are standing-bearing only when quotient structure and witnesses persist; and why threshold crossing requires boundary trace.

The strongest defensible conclusion is therefore:

The runnability of a parameter is determined not by whether it is numerical, dimensionless, or empirically fitted, but by whether variation preserves standing-bearing identity through admissible transport. Running is a role-dependent structural property, not a generic freedom of parameters.

A Theorem Ladder and Claim Index

Table 5: Main definitions and results.

Item	Role in the manuscript
Definition 3.1	Establishes the same-domain comparison target.
Definition 3.2	Introduces scale-indexed domains and RG transport maps.
Definition 3.3	Specifies the common empirical reconstruction domain.
Definition 3.4	Defines physical parameter identity by standing-bearing use.
Definition 3.5	Defines genuine running as admissible transport.
Lemma 4.1	Separates presentation change from physical running.
Lemma 4.2	Defines effective drift.
Lemma 4.3	Defines transport-governed running.
Lemma 4.4	Defines quotient-stable invariance.
Lemma 4.5	Establishes anchor non-runnability.

Item	Role in the manuscript
Theorem 5.2	Proves the exhaustion of parameter mobility modes.
Proposition 6.1	Shows dimensionless status does not determine mobility.
Proposition 7.1	Explains why couplings run.
Proposition 8.1	Explains why gauge structure does not run same-domain.
Proposition 12.2	Specifies cross-threshold mobility through boundary trace.
Corollary 13.2	Blocks null endpoints as physical completions without standing.

B Glossary

Admissibility envelope The regime in which a description licenses physical reference and comparison.

Boundary trace The overlap information at a regime transition that permits downstream reconstruction.

Effective drift Envelope-bound parameter change licensed by matching, fitting, or regime validity.

Overlap witnesses Observables or reconstruction relations shared by two descriptions after admissible comparison.

Presentation change Variation of coordinate, scheme, unit, or representation without change in standing-bearing content.

Quotient-stable invariance Persistence of invariant content under admissible identification despite raw or microscopic differences.

Standing-bearing identity Physical identity preserved through empirical reconstruction, lawful transport, quotient structure, or envelope continuity.

Transport-defined running Parameter mobility governed by lawful scale transport preserving overlap witnesses and quotient identity.

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